

CLUSTERING

PROXIMITY – PART I

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PROXIMITY

The following concepts will be introduced:

- ✓ **SIMILARITY**
- ✓ **DISSIMILARITY**
- ✓ **PROXIMITY**
- ✓ **SIMILARITY/DISSIMILARITY FOR SINGLE ATTRIBUTE RECORDS**
 - NOMINAL
 - ORDINAL
 - NUMERIC

CLUSTER ANALYSIS is rooted into the concepts of **SIMILARITY** and **DISSIMILARITY**, and in many cases, once the similarities and dissimilarities have been computed, the available data set is not needed anymore.

The **SIMILARITY** between two objects (records) is a numerical measure of the degree to which the two objects (records) are alike. Similarities are

- ✓ higher for pairs of objects (records) that are more alike
- ✓ usually non-negative and are often between 0 (no similarity) and 1 (complete similarity)



The **DISSIMILARITY** between two objects (records) is a numerical measure of the degree to which the two objects (records) are different.

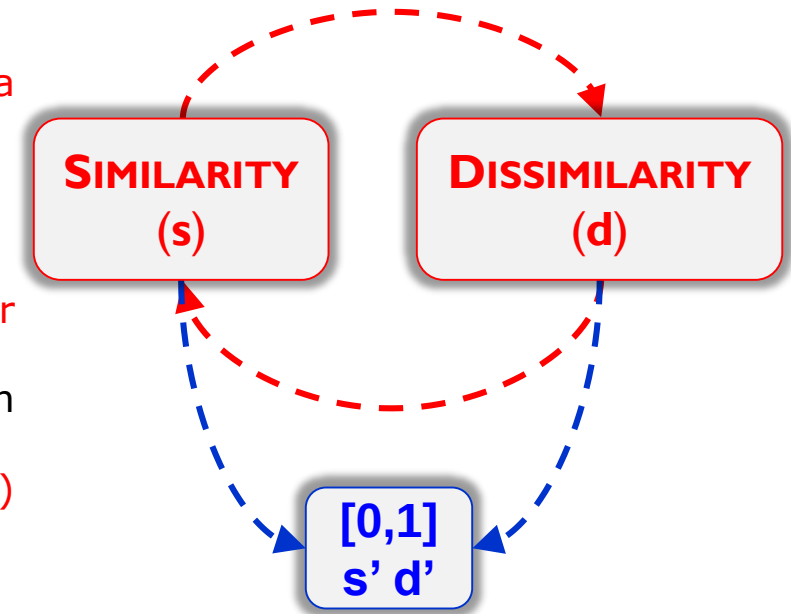


DISSIMILARITIES sometimes fall in the interval $[0,1]$, but it is also common for them to take values between 0 and ∞ . Dissimilarities are lower for more similar pairs of objects.

For convenience, the term **PROXIMITY** will be used to refer to either **SIMILARITY** or **DISSIMILARITY**.

TRANSFORMATIONS are often applied to convert a similarity (**s**) to a dissimilarity (**d**) or vice versa.

Frequently, **PROXIMITIES**, especially similarities, are defined or transformed to take values on $[0,1]$, in such a way to use a scale in which proximity indicates the fraction of similarity (dissimilarity) between two records.



If the similarities (**s**) between records take value between 1 (not at all similar) and 10 (completely similar), we can make new similarities values (**s'**) to fall into the $[0,1]$ interval by using the following transformation:

$$s' = \frac{s-1}{9}$$

In general, the new similarity **s'** and the new dissimilarity **d'** are defined as follows

$$s' = \frac{s - \min_s}{\max_s - \min_s}$$

$$d' = \frac{d - \min_d}{\max_d - \min_d}$$

There are many complications in mapping proximity measures to $[0,1]$.

If the proximity measure takes values on $[0,\infty]$, then a non-linear transformation is needed and values will not have the same relationship to one another on the new scale.

Consider the transformation

$$d' = \frac{d}{1+d}$$

for a dissimilarity measure (d) taking values on $[0,\infty]$, you can have the following effect

DISSIMILARITY (d)		DISSIMILARITY (d')
0		0.000
0.5		0.333
2		0.667
10	$\xrightarrow{\quad d' = \frac{d}{1+d} \quad}$	0.900
100	$\xrightarrow{\quad \text{NON-LINEAR TRANSFORMATION} \quad}$	0.990
1000		0.999



Larger values of the original dissimilarity (d) scale are compressed into the range of values near 1 for d'

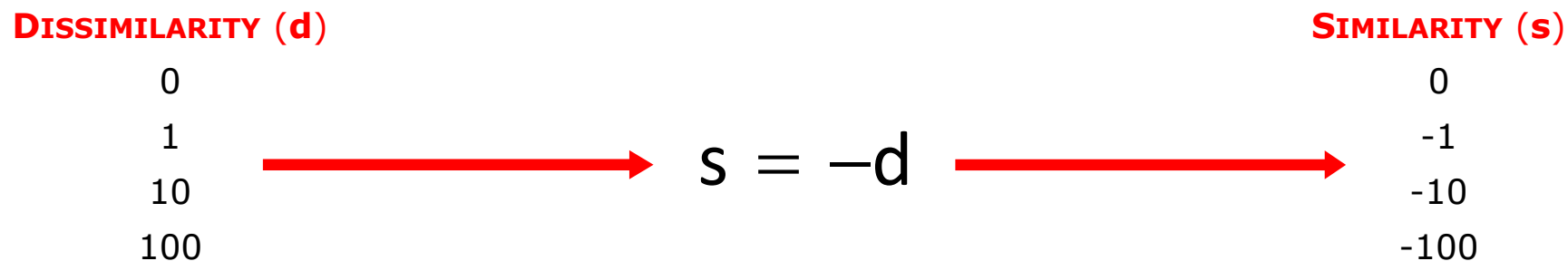
Whether or not this is desirable depends on the application

Transforming similarities to dissimilarities and vice versa is also relatively straightforward, although we again face the issues of preserving meaning and changing a linear scale into a non-linear scale.

If the similarity (**s**) (dissimilarity (**d**)) falls in $[0,1]$, then the dissimilarity (**d**) (similarity (**s**)) can be defined as follows

$$d = 1 - s \quad (s = 1 - d)$$

Another approach is to define similarity (**s**) as the negative of the dissimilarity (**d**) (or vice versa).



In general any monotonic decreasing function can be used to transform similarities (**s**) to dissimilarities (**d**), or vice versa.

However, the **PROXIMITY** between two records (data objects) is a function of the proximity between the corresponding attributes of the two records.

Day Charge
45.07
27.47

We first describe how to measure the proximity between records having only one simple attribute, then the case of records with multiple attributes is considered.

We now discuss **PROXIMITY** between **RECORDS HAVING THE SAME SINGLE ATTRIBUTE**.

ATTRIBUTE TYPE		DISSIMILARITY	SIMILARITY
CATEGORICAL (QUALITATIVE)	NOMINAL	$d = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{otherwise} \end{cases}$	$s = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise} \end{cases}$
	ORDINAL	$d = \frac{ x - y }{n - 1}$	$s = 1 - d$
NUMERIC (QUANTITATIVE)	INTERVAL	$d = x - y $	$s = -d \quad s = \exp(-d)$
	RATIO		$s = 1 - \frac{d - \min_d}{\max_d - \min_d}$

All we can say is that **two records** either have the same value or they do not.

ATTRIBUTE TYPE		DISSIMILARITY	SIMILARITY
(QUALITATIVE)	NOMINAL	$d = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{otherwise} \end{cases}$	$s = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise} \end{cases}$

Consider **TWO RECORDS** consisting of the **SAME SINGLE NOMINAL ATTRIBUTE**.



BINARY $\longrightarrow$ The **GENDER** of a person can be $\{male, female\}$

The **EYES COLOR** of a person can be $\{amber, brown, green, blue, gray\}$



WHAT WOULD IT MEAN FOR TWO SUCH RECORDS TO BE SIMILAR?

Consider **TWO RECORDS** consisting of the **SAME SINGLE ORDINAL ATTRIBUTE**.



THE SITUATION IS MORE COMPLICATED BECAUSE INFORMATION ABOUT ORDER SHOULD BE TAKEN INTO ACCOUNT

CATEGORICAL	ORDINAL	$d = \frac{ x - y }{n - 1}$	$s = 1 - d$

The **QUALITY** of a product can be *{poor, fair, OK, good, wonderful}*

Quality(P1) = *wonderful*

Quality(P2) = *good*

Quality(P3) = *OK*

It would seem reasonable that **P1** would be **closer to P2** than it would be **to P3**

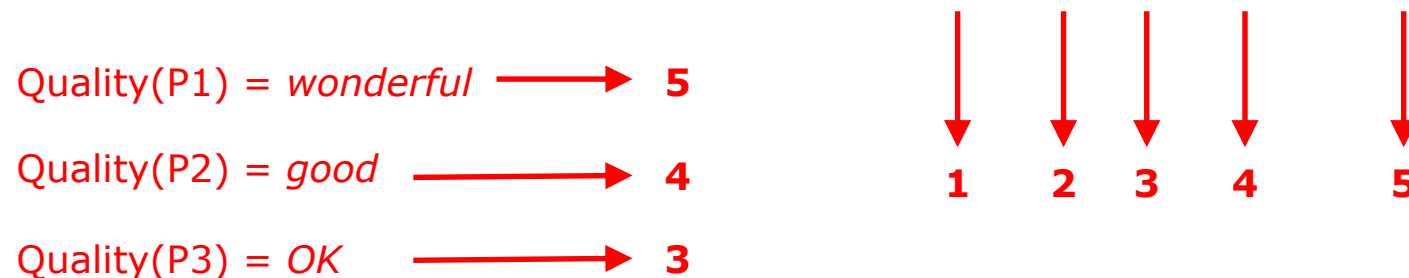
Consider **TWO RECORDS** consisting of the **SAME SINGLE ORDINAL ATTRIBUTE**.



THE SITUATION IS MORE COMPLICATED BECAUSE INFORMATION ABOUT ORDER SHOULD BE TAKEN INTO ACCOUNT

CATEGORICAL	ORDINAL	$d = \frac{ x - y }{n - 1}$	$s = 1 - d$

The **QUALITY** of a product can be *{poor, fair, OK, good, wonderful}*



Consider **TWO RECORDS** consisting of the **SAME SINGLE ORDINAL ATTRIBUTE**.



THE SITUATION IS MORE COMPLICATED BECAUSE INFORMATION ABOUT ORDER SHOULD BE TAKEN INTO ACCOUNT

CATEGORICAL	ORDINAL	$d = \frac{ x - y }{n - 1}$	$s = 1 - d$

The **QUALITY** of a product can be {*poor, fair, OK, good, wonderful*}

Quality(P1) = *wonderful* $\longrightarrow$ **5**

$| \text{Quality(P1)} - \text{Quality(P2)} | = 5 - 4 = 1$

Quality(P2) = *good* $\longrightarrow$ **4**

$d(\text{P1}, \text{P2}) = (5 - 4) / 4 = 0.25$

Quality(P3) = *OK* $\longrightarrow$ **3**

$d(\text{P2}, \text{P3}) = (4 - 3) / 4 = 0.25$



**You are assuming
equal intervals**

Consider **TWO RECORDS** consisting of the **SAME SINGLE INTERVAL OR RATIO ATTRIBUTE**.



THE NATURAL MEASURE OF DISSIMILARITY BETWEEN TWO RECORDS IS THE ABSOLUTE DIFFERENCE OF THEIR VALUES

Comparing the value of the **INTL CHARGE** of two **customers** of the **Telecom Company**.

The **dissimilarities** take values **between 0 and infinite**, rather than between 0 and 1.

NUMERIC (QUANTITATIVE)	INTERVAL	$d = x - y $	$s = -d$	$s = \exp(-d)$
	RATIO		$s = \frac{1}{1 + d}$	$s = 1 - \frac{d - \min_d}{\max_d - \min_d}$